

Exercise sheet N°3: Complex numbers

Exercise 1

1. Write the following complex numbers in algebraic form $x + iy$.

$$A = \frac{2+3i}{1-2i} \quad B = \frac{1+i}{1-i} \quad C = \frac{(1+i)^5}{(1-i)^3} \quad D = \left(\frac{1+i}{1-i}\right)^3 + \frac{2+3i}{1-2i}$$

2. Using the polar form show that: $e^{i2\pi} = 1$, $e^{i\pi} = -1$, $e^{i\frac{\pi}{2}} = i$ and $e^{i\frac{-\pi}{2}} = -i$.

3. Deduce that: (a) $\forall k \in \mathbb{Z}, e^{i(2k\pi)} = 1$ (b) $\forall k \in \mathbb{Z}, e^{i(2k+1)\pi} = -1$ (c) $e^{i\frac{35\pi}{12}} = e^{i\frac{11\pi}{12}}$.

Exercise 2

Let $z = -1 - i$ and $w = 1 - i\sqrt{3}$ be two complex numbers.

- Determine the conjugate and the modulus to each z and w , and then determine an argument to each z and w in $[0, 2\pi]$.
- Express z and w in polar and exponential forms.
- Determine the polar and exponential forms of zw .
- Deduce $\cos \frac{11\pi}{12}$ and $\sin \frac{11\pi}{12}$.
- Using De Moivre's formula, prove that $z^4 = -4$ and $z^{-3} = \frac{-1}{8}$.

Exercise 3

Solve the following equations in $\mathbb{C}$.

A.

- (A1) $z - 2 = i(2z + 3)$
- (A2) $z^4 - 1 = 0$
- (A3) $(z - 1)^3 - i = 0$
- (A4) $z^2 = 2z - 4$
- (A5) $z^2 - (1 + 2i)z + i - 1 = 0$

B.

- (B1) $z - 1 = i(z + 1)$
- (B2) $z^4 + 2 = 0$
- (B3) $(z + 1)^3 + i = 0$
- (B4) $z^2 = -z - 1$
- (B5) $z^2 - (2 - i)z + 1 - i = 0$

Exercise 4 (Final Exam 24/25)

Let $k = \frac{2i+2\sqrt{3}}{4i}$ be a complex number.

- Write k in algebraic form $(x + iy)$.
- Determine the modulus r and an argument θ in $[0, 2\pi]$ for k , then write k in exponential form.
- Using De Moivre's formula, find the real number k^3 .
- Solve in $\mathbb{C}$ the equation $z^3 - k^3 = 0$.